

10.1 The Manifold

No change. The manifold $H^2 \times SPD(3)$ stays.

10.2 Why Geometry Enforces Safety

No change.

10.3 The Safety Threshold Λ

Delete equations 10.1, 10.2,

KEEP 10.3 as currently written.

Replace with:

The threshold Λ is derived from the truncated Euler product over primes $\{2, 3, 5, 7, 11, 13\}$:

$$I = \prod_{p \leq 13} (1 - p^{-1/2}) = 0.0214857126 \quad (10.1)$$

$$\Lambda = 1 - I = 0.9785142874 \quad (10.2)$$

The conservation law $I + \Lambda = 1$ holds exactly. Λ is the spectral complement of the Euler attenuation product over the pure kernel.

Crucially, Λ is never hardcoded. It is computed dynamically at every initialization via the Sieve of Eratosthenes, making it mathematically forced by the prime numbers rather than empirically tuned.

10.4 The Decision Metric SROI

No change to equations 10.4 and 10.5. The geodesic distance d_M and the SROI mapping $SROI = \exp(-d_M/50)$ stand. But note in the surrounding text that Λ now equals 0.9785142874, not 0.9583.

10.5 The Decision Rule

Equation 10.6 stays as written, with $\Lambda = 0.9785142874$:

$$Decision = \begin{cases} ACCEPT & \text{if } SROI \geq \Lambda \\ REJECT & \text{if } SROI < \Lambda \end{cases} \quad (10.6)$$

10.6 Zero Safety Violations

No change.

10.7 Deterministic Auditability

No change. The audit hash formula stands:

$$Hash = SHA256(input || decision || SROI || \Lambda) \quad (10.7)$$

Summary of edits

Location	Before	After
10.3	$I = 0.021486$	$I = 0.0214857126$
10.3	$K = \ L_{13}\ /I = 44.601732$	deleted
10.3	$\Lambda = I \times K = 0.9583$	$\Lambda = 1 - I = 0.9785142874$
10.5	$\Lambda = 0.9583$ implicit	$\Lambda = 0.9785142874$
10.7	hash over $\Lambda = 0.9583$	hash over $\Lambda = 0.9785142874$

Everything else in Chapter 10 is unaffected. The manifold, the geodesic distance, the SROI formula, the decision rule, the hard-stop, the zero-violations claim, and the audit hash structure all remain valid. Only the constant and its derivation change.

One consistency check

If Chapter 10 uses $SROI \geq \Lambda$ with $\Lambda = 0.9785142874$, and the H2E Sheriff V3 notebook uses $SROI = \text{clamp}(\cos(H \cdot z, w) \times \Lambda \times 1.043, 0, 1)$ with the same Λ , then the effective cosine threshold in both is $1/1.043 \approx 0.9588$. That should be stated once in Chapter 10 so the two formulations are visibly reconciled: the manifold gate uses Λ directly, and the vector-space gate uses $\Lambda \times 1.043$ which produces the same threshold when converted to a cosine requirement.